

- 17** A red indicator solution was mixed with agar and the resulting solid was cut into small cylindrical blocks. The blocks were placed in an acid which turns the indicator yellow and all other variables were kept constant. The dimensions of the blocks are shown.

block 1 height 3 mm diameter 6 mm

block 2 height 6 mm diameter 12 mm

block 3 height 8 mm diameter 16 mm

The formula for calculating the surface area of a cylinder is $2\pi rh + 2\pi r^2$. The formula for calculating the volume of a cylinder is $\pi r^2 h$.

Which row shows the correct surface area (SA) to volume (V) ratio for each block and the time taken for the block to turn yellow?

	block 1		block 2		block 3	
	SA to V ratio	time to turn yellow / mins	SA to V ratio	time to turn yellow / mins	SA to V ratio	time to turn yellow / mins
A	0.75 : 1.0	4	1.5 : 1.0	5	2.0 : 1.0	11
B	0.75 : 1.0	11	1.5 : 1.0	5	2.0 : 1.0	4
C	1.33 : 1.0	4	0.67 : 1.0	5	0.5 : 1.0	11
D	1.33 : 1.0	11	0.67 : 1.0	5	0.5 : 1.0	4